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MTE 204 Spring 2026

Numerical Methods

Section 001, 101

Class Schedule

Course	Meet Days	Meet Time	Location	Instructor(s)
MTE 204 001 [LEC]	Tue, Thu <i>May 11 - Aug 5</i>	03:00PM - 04:20PM	E7 3353	Duane Cronin duane.cronin@uwaterloo.ca
MTE 204 101 [TUT]	Fridays <i>May 11 - Aug 5</i>	10:30AM - 11:20AM	E7 3353	

schedule data automatically refreshed daily

Instructional Team

Course Instructor

Duane Cronin dscronin@uwaterloo.ca

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Teaching Assistants

Sophia Ngan s3ngan@uwaterloo.caYiren Xu yiren.xu@uwaterloo.ca

Course Description

Calendar Description for MTE 204

Number systems and machine errors; roots of non-linear equations; matrix calculations; Eigenvalue and Eigenvector calculations; interpolation and approximation; numerical integration and solution of ODE's (linear and non-linear) and systems of ODEs; calculation of series; solution methods for PDE's; use of numeric and symbolic computing tools.

[View requirements for MTE 204 ^{\(1\)}](#)

Projects

Two small projects will be assigned during the term, one before and one after the midterm, to demonstrate engineering-based applications of numerical solution methods. These projects will require the use of MatLab. All project work must be

completed on an individual basis.

Problem Sets

Problem sets and solutions will be posted on the course website. Students are expected to work independently on these problem sets according to the course schedule and check their work using the solutions provided.

Learning Outcomes

By the end of this course students should be able to...

1. Develop and implement stable and accurate numerical methods to solve linear systems of equations, find root of equations and perform curve fitting, numerical integration and differentiation [GA: KB (I)]
 2. Understand the basic concepts of partial differential equations, including initial and boundary conditions and analytical solution techniques [GA: KB (I)]
 3. Use numerical methods, including finite difference and finite element analysis, to solve ordinary differential equations and partial differential equations for engineering problems [GA: KB (I)]
- (GA: Graduate Attributes, KB: Knowledge Base)

Tentative Class Plan

Week	Date	Topic	Text Book Sections	Problem Sets
1	11-May	Introduction, Error definition	3.1-3.4, 4.1-4.4	
		Roots of Equations - Graphical methods	5.1	
2	18-May	Roots of Equations - Bisection method	5.2	
		Root finding, Open methods - Newton Raphson	6.1, 6.2	
		Secant Method	6.3	PS #1
3	25-May	Systems of Linear Equations	PT3.1-3.3, 9.1	
		Gaussian Elimination, Pivoting and Scaling	9.3, 9.4	PS#2
		Iterative methods, Gauss Seidel	11.1, 11.2	
4	01-Jun	Gauss Seidel with relaxation	11.1, 11.2	PS#3
		Curve Fitting, Polynomial Regression	17.1, 17.2	PS#4
5	08-Jun	Polynomial and Spline Interpolation	18.1	
		Spline Interpolation	18.1, 18.2, 18.6	
6	15-Jun	Numerical Integration, Romberg Integration	21.1, 21.2, 22.2	
		Gaussian Quadrature	22.4	
		Numerical Differentiation	23.1, 23.2 + 4.1	
7	22-Jun	Mid-Term Week, MTE 204 Mid-term test Tuesday Oct 24, 11:30am-1:30pm		
8	29-Jun	Numerical Solution of Ordinary Differential Equations (ODEs)	25.1	
		Euler's Method	25.2, 26.1	

9	06-Jul	Runge Kutta Method	25.3	
		Higher Order Runge Kutta Methods	25.3, 25.4	
		Boundary Value Problems	27.1	PS#5
10	13-Jul	Finite Element Analysis (FEA) Introduction	31.1	
		FEA, 1D Analysis	31.2	
		FEA, 1D Spring and 2D Elements	31.2	PS#6
11	20-Jul	Introduction, Partial Differential Equations (PDEs)	PT8.1, PT8.2	
		Laplace and Wave Equations		
		Boundary and Initial Conditions, PDE Solutions		
12	27-Jul	Separation of Variables Solution, PDEs		PS#7
		Numerical Solution of PDEs	29.1, 29.2	
		Finite Difference Solution of Parabolic Equations	30.1, 30.2	
13	03-Aug	Finite Difference Solution of 1D Diffusion Equation	30.3, 30.4	
	+Tues Aug 4	Finite Difference Solution of 1D Wave Equation		PS#8

Required Materials & Technologies

Note: Any prices provided in course outlines are best estimates based on recent online prices and do not include shipping or taxes. Prices may vary between retailers.

This course has additional costs to students (listed in sections below).

Readings

Title / Name	Notes / Comments	Required	Used Versions Allowed	Price (CAD)
Steven C. Chapra and Raymond P. Canale, Numerical Methods for Engineers, 8th ed. McGraw Hill	ISBN: 978-1-260-57138-7 (this is the International Student Edition)	Recommended	Yes	109.95

Print (Publisher) : ISBN 9781260571387

⁽²⁾<https://www.mheducation.ca/pro...> ⁽³⁾

Price : \$109.95

Digital (Vitalsource) : ISBN 9781260590401

⁽⁴⁾<https://uwaterloo-store.vitals...> ⁽⁵⁾

180-day subscription - \$59.00

Lifetime subscription - \$99.00

Technology

Name of Technology	Notes / Comments	Required	URL (student access)	Price (CAD)
MatLab	Student edition available through UW	Required		0.00
Excel or other spreadsheet software	Student software available through UW	Required		0.00

Assessments & Activities

Component / Activity	Date or Due Date	Location / Submission Method	Weight (%)
Midterm	Tuesday June 23, 9:30am	in-person, PSE 3343 and PSE 3353	35%
Project #1	Prior to midterm, timing tbd	individual project, submitted via Learn	10%
Project #2	After midterm, timing tbd	individual project, submitted via Learn	10%
Final Exam	tbd	in person, location tbd	45%

Exams

The midterm and final exams will be conducted in person. Exams are closed-book, students will be allowed a single page formula sheet.

Projects

Two projects will be assigned during the term, one before and one after the midterm, to demonstrate engineering-based applications of numerical solution methods. These projects will require the use of MatLab. All project work must be completed on an individual basis.

Late / Missed Content

Late penalty for graded activities (projects) is 10% per day.

Assignment Screening

No automated assignment screening will be utilized within this course.

Generative AI

Generative artificial intelligence (GenAI) trained using large language models (LLM) or other methods to produce text, images, music, or code, like Chat GPT, DALL-E, or GitHub CoPilot, **may be used under specific conditions** in this course with proper documentation, citation, and acknowledgement. **Permitted uses of and expectations for using GenAI will discussed in class and outlined on assignment instructions.**

[Recommendations for how to cite generative AI in student work at the University of Waterloo may be found through the Library.](#) ⁽⁶⁾

Please be aware that generative AI is known to falsify references to other work and may fabricate facts and inaccurately express ideas. GenAI generates content based on the input of other human authors and may therefore contain inaccuracies or reflect biases.

To protect the privacy and security of any data entered, students should use the University's version of [Co-Pilot](#) ⁽⁷⁾ and login with their UW ID. Data entered into other systems can be added to training sets, monitored, geolocated and even reproduced as output which may share private personal information or result in intellectual property breaches.

In addition, you should be aware that the legal/copyright status of generative AI inputs and outputs is unclear. Exercise caution when using large portions of content from AI sources, especially images. [More information is available from the Copyright Advisory Committee.](#) ⁽⁸⁾

You are accountable for the content and accuracy of all work you submit in this class, including any supported by generative AI. You should be able to readily demonstrate your knowledge of your submissions. To demonstrate your learning, you should keep your rough notes, including sources, research notes, brainstorming, drafting notes and prompts. You may be asked to submit these notes along with earlier drafts of your work, either through saved drafts or saved versions of a document.

Administrative Policy

Faculty of Engineering Guiding Practices

Territorial Acknowledgement: The University of Waterloo acknowledges that much of our work takes place on the traditional territory of the Neutral, Anishinaabeg and Haudenosaunee peoples. Our main campus is situated on the Haldimand Tract, the land granted to the Six Nations that includes six miles on each side of the Grand River. Our active work toward reconciliation takes place across our campuses through research, learning, teaching, and community building, and is centralized within the [Office of Indigenous Relations](#) ⁽⁹⁾.

Inclusive Teaching-Learning Spaces: The University of Waterloo values the diverse and intersectional identities of its students, faculty, and staff. The University regards equity and diversity as an integral part of academic excellence and is committed to accessibility for all. We consider our classrooms, online learning, and community spaces to be places where we all will be treated with respect, dignity, and consideration. We welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability – and other visible and nonvisible differences. We are all expected to contribute to a respectful, welcoming, and inclusive teaching- learning environment. Any member of the campus community who has experienced discrimination at the University is encouraged to seek guidance from the [Office of Equity, Diversity, Inclusion & Anti-racism \(EDI-R\)](#) ⁽¹⁰⁾ via email at equity@uwaterloo.ca. [Sexual Violence Prevention & Response Office \(SVPRO\)](#) ⁽¹¹⁾, supports students at UWaterloo who have experienced, or have been impacted by, sexual violence and gender-based violence. This includes those who experienced harm, those who are supporting others who experienced harm. SVPRO can be contacted at svpro@uwaterloo.ca

Religious & Spiritual Observances: The University of Waterloo has a duty to accommodate religious and spiritual observances under the Ontario Human Rights Code. Please inform the instructor at the beginning of term if special accommodation needs to be made for religious observances that are not otherwise accounted for in the scheduling of classes and assignments. Consult with your instructor(s) within two weeks of the announcement of the due date for which accommodation is being sought.

Respectful Communication and Pronouns: Communications with Instructor(s) and teaching assistants (TAs) should be through recommended channels for the course (e.g., email, LEARN, Piazza, Teams, etc.) Please use your UWaterloo email

address. Include an academic signature with your full name, program, student ID. We encourage you to include your pronouns to facilitate respectful communication (e.g., he/him; she/her; they/them). You can update your chosen/preferred name at [WatIAM](#).⁽¹²⁾ You can update your pronouns in [Quest](#).⁽¹³⁾

Mental Health and Wellbeing Resources: If you are facing challenges impacting one or more courses, contact your academic advisor, Associate Chair Undergraduate, or the Director of your academic program. Mental health is a serious issue for everyone and can affect your ability to do your best work. We encourage you to seek out mental health and wellbeing support when needed. The [Faculty of Engineering Wellness](#) [Program](#)⁽¹⁴⁾ has programming and resources for undergraduate students. For counselling (individual or group) reach out to [Campus Wellness and Counselling Services](#).⁽¹⁶⁾ Counselling Services is an inclusive, non-judgmental, and confidential space for anyone to seek support. They offer confidential counselling for a variety of areas including anxiety, stress management, depression, grief, substance use, sexuality, relationship issues, and much more.

Intellectual Property: Be aware that this course contains the intellectual property of their instructor, TA, and/or the University of Waterloo. Intellectual property includes items such as:

- Lecture content, spoken and written (and any audio/video recording thereof).
- Lecture handouts, presentations, and other materials prepared for the course (e.g., PowerPoint slides).
- Questions or solution sets from various types of assessments (e.g., assignments, quizzes, tests, final exams); and
- Work protected by copyright (e.g., any work authored by the instructor or TA or used by the instructor or TA with permission of the copyright owner).

Course materials and the intellectual property contained therein are used to enhance a student's educational experience. However, sharing this intellectual property without the intellectual property owner's permission is a violation of intellectual property rights. For this reason, it is necessary to ask the instructor, TA and/or the University of Waterloo for permission before uploading and sharing the intellectual property of others online (e.g., to an online repository).

Permission from an instructor, TA or the University is also necessary before sharing the intellectual property of others from completed courses with students taking the same/similar courses in subsequent terms/years. In many cases, instructors might be happy to allow distribution of certain materials. However, doing so without expressed permission is considered a violation of intellectual property rights and academic integrity.

Please alert the instructor if you become aware of intellectual property belonging to others (past or present) circulating, either through the student body or online.

Continuity Plan - Fair Contingencies for Unforeseen Circumstances (e.g., resurgence of COVID-19): In the event of emergencies or highly unusual circumstances, the instructor will collaborate with the Department/Faculty to find reasonable and fair solutions that respect rights and workloads of students, staff, and faculty. This may include modifying content delivery, course topics and/or assessments and/or weight and/or deadlines with due and fair notice to students. Substantial changes after the first week of classes require the approval of the Associate Dean, Undergraduate Studies.

Declaring absences: *[undergraduate students and/or courses only]* Regardless of the process used to [declare an absence](#) [\(17\)](#), students are responsible for reaching out to their instructors as soon as possible. The course instructor will determine how missed course components are accommodated. Self-declared absences are submitted through [Quest](#) [\(18\)](#). Requests for absences requiring documentation (e.g., Verification of Illness Form, bereavement, etc.) are submitted on the [Absence Database](#) [\(19\)](#). The [UWaterloo Verification of Illness form](#) [\(20\)](#), completed by a health professional, is the preferred documentation for an absence due to illness. Do not send medical documentation to your advisor, course instructor, teaching assistant, or lab coordinator. Once approved, your instructor will be notified of your absence.

Rescheduling Co-op Interviews: Follow the co-op process for [rescheduling co-op interviews](#) ⁽²¹⁾ for conflicts with important assessments (e.g., presentations, laboratories, tests, and exams). Attendance at co-operative work-term employment interviews is not considered to be a valid reason to miss a graded course component.

Accreditation Data Collection

The Faculty of Engineering may be collecting information from this course for accreditation purposes, including course outline, exams, assignments, and samples of student work. The [Canadian Engineering Accreditation Board \(CEAB\)](#) ⁽²²⁾ periodically reviews the Bachelor of Applied Science (BASc) and the Bachelor of Software Engineering (BSE) degrees. The [Canadian Architectural Certification Board \(CACB\)](#) ⁽²³⁾ periodically reviews the Bachelor of Architectural Studies (BAS) degree and the Master of Architecture (MArch) degree.

University Policy

Mental Health: At the University of Waterloo, we are dedicated to supporting your mental and emotional well-being. Our Counselling Services offer confidential support, including individual counselling, workshops, and crisis intervention. If you're struggling, please reach out for help at 519-888-4096 or visit [their website](#) ⁽²⁴⁾ for more information.

Academic integrity: In order to maintain a culture of academic integrity, members of the University of Waterloo community are expected to promote honesty, trust, fairness, respect and responsibility.

Grievance: A student who believes that a decision affecting some aspect of their university life has been unfair or unreasonable may have grounds for initiating a grievance. Read [Policy 70, Student Petitions and Grievances, Section 4](#) ⁽²⁵⁾. When in doubt, please be certain to contact the department's administrative assistant who will provide further assistance.

Discipline: A student is expected to know what constitutes academic integrity to avoid committing an academic offence, and to take responsibility for their actions. A student who is unsure whether an action constitutes an offence, or who needs help in learning how to avoid offences (e.g., plagiarism, cheating) or about "rules" for group work/collaboration should seek guidance from the course instructor, academic advisor, or the undergraduate associate dean. For information on categories of offences and types of penalties, students should refer to [Policy 71, Student Discipline](#) ⁽²⁶⁾. For typical penalties, check [Guidelines for the Assessment of Penalties](#) ⁽²⁷⁾.

Appeals: A decision made or penalty imposed under [Policy 70, Student Petitions and Grievances](#) ⁽²⁸⁾ (other than a petition) or [Policy 71, Student Discipline](#) ⁽²⁹⁾ may be appealed if there is a ground. A student who believes they have a ground for an appeal should refer to [Policy 72, Student Appeals](#) ⁽³⁰⁾.

Note for students with disabilities and disabling conditions: The University of Waterloo recognizes its obligations under the Ontario Human Rights Code to accommodate students with known or suspected disabilities and disabling conditions (e.g. medical conditions, injuries, impacts of trauma such as from violence or discrimination) to the point of undue hardship. To support this obligation, [AccessAbility Services](#) ⁽³¹⁾ (AAS) collaborates with all academic departments and schools to facilitate academic accommodations for students with disabilities and disabling conditions without compromising the academic integrity of the curriculum. If you believe you may require academic accommodations (e.g., testing accommodations, classroom accommodations), register with AAS as early in the term as possible by completing the [online application](#) ⁽³²⁾. Students already registered with AAS must activate their accommodations for each of their courses at the beginning of each term using AAS' online system. If you require assistance, contact AAS by phone (519-888-4567 ext. 35082), email (access@uwaterloo.ca) or in-person (Needles Hall North, 1st Floor, Room 1401).

Turnitin.com: Text matching software (Turnitin®) may be used to screen assignments in this course. Turnitin® is used to verify that all materials and sources in assignments are documented. Students' submissions are stored on a U.S. server,

therefore students must be given an alternative (e.g., scaffolded assignment or annotated bibliography), if they are concerned about their privacy and/or security. Students will be given due notice, in the first week of the term and/or at the time assignment details are provided, about arrangements and alternatives for the use of Turnitin in this course.

It is the responsibility of the student to notify the instructor if they, in the first week of term or at the time assignment details are provided, wish to submit alternate assignment.

Reference: Links from Document

1. <https://acal.fast.uwaterloo.ca/course/1265/MTE/204>
2. <https://www.mheducation.ca/product/ise-numerical-methods-for-engineers-9781260571387-can-group>
3. <https://www.mheducation.ca/product/ise-numerical-methods-for-engineers-9781260571387-can-group>
4. <https://uwaterloo-store.vitalsource.com/products/ise-ebook-online-access-for-numerical-methods-for-chapra-v9781260590401>
5. <https://uwaterloo-store.vitalsource.com/products/ise-ebook-online-access-for-numerical-methods-for-chapra-v9781260590401>
6. https://subjectguides.uwaterloo.ca/chatgpt_generative_ai
7. <https://copilot.microsoft.com/>
8. <https://uwaterloo.ca/copyright-at-waterloo/teaching/generative-artificial-intelligence>
9. <https://uwaterloo.ca/indigenous>
10. <https://uwaterloo.ca/equity-diversity-inclusion-anti-racism/>
11. <https://uwaterloo.ca/sexual-violence-prevention-response-office>
12. <https://idm.uwaterloo.ca/watiam/>
13. <https://uwaterloo.ca/quest/help/students/how-do-i/view-or-update-my-personal-information>
14. <https://uwaterloo.ca/engineering-wellness-program/>
15. <https://uwaterloo.ca/engineering-wellness-program/>
16. <https://uwaterloo.ca/campus-wellness/counselling-services>
17. <https://uwaterloo.ca/engineering/undergraduate-students/getting-support/declaring-absences>
18. <https://uwaterloo.ca/quest/help/students/how-do-i/self-declare-absence-undergraduate-students>
19. <https://vif.uwaterloo.ca/>
20. <https://uwaterloo.ca/campus-wellness/health-services/student-medical-clinic/verification-illness-services>
21. <https://uwaterloo.ca/co-operative-education/find-your-co-op-job/find-job-waterlooworks/interview/interview-conflicts>
22. <https://engineerscanada.ca/accreditation/accreditation-board>
23. <https://cacb.ca/>
24. <https://uwaterloo.ca/students/health-and-well-being/counselling-appointments>
25. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-70>
26. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-71>
27. <https://uwaterloo.ca/secretariat/guidelines/guidelines-assessment-penalties>
28. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-70>
29. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-71>
30. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-72>
31. <https://uwaterloo.ca/accessability-services/>
32. <https://uwaterloo.ca/accessability-services/students/applying-academic-accommodations/documentation-information-forms>

For technical support please email OutlineSupport@uwaterloo.ca

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